

Product Carbon Footprint Study Report

Product Name: keyboard

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 29, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: keyboard

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1Item keyboard

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1Item keyboard.

2. System Boundary

keyboard production, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of keyboard produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **30.066 kg CO₂-Eq.**

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

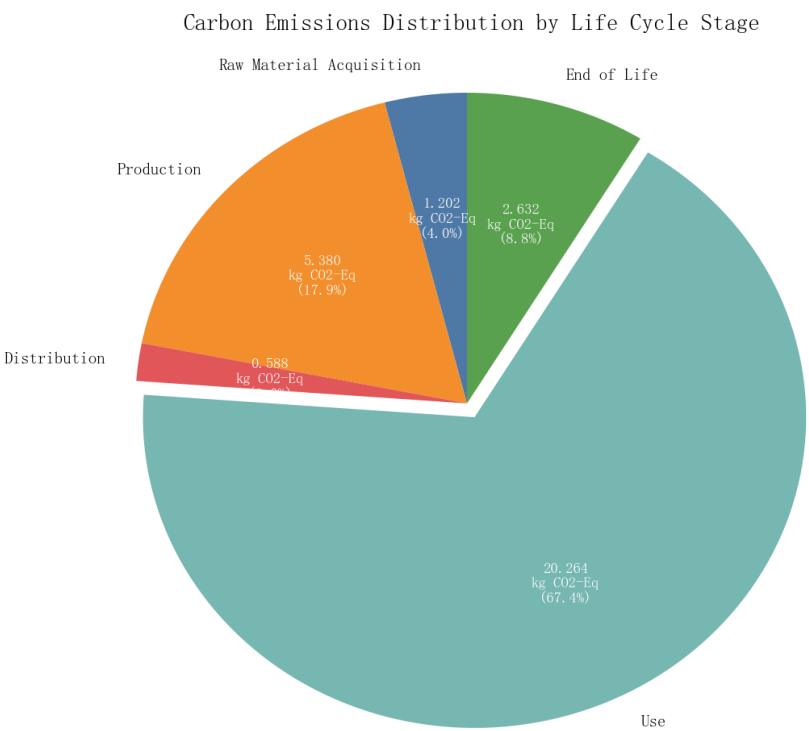


Table 1: Carbon Emissions by Life Cycle Stage

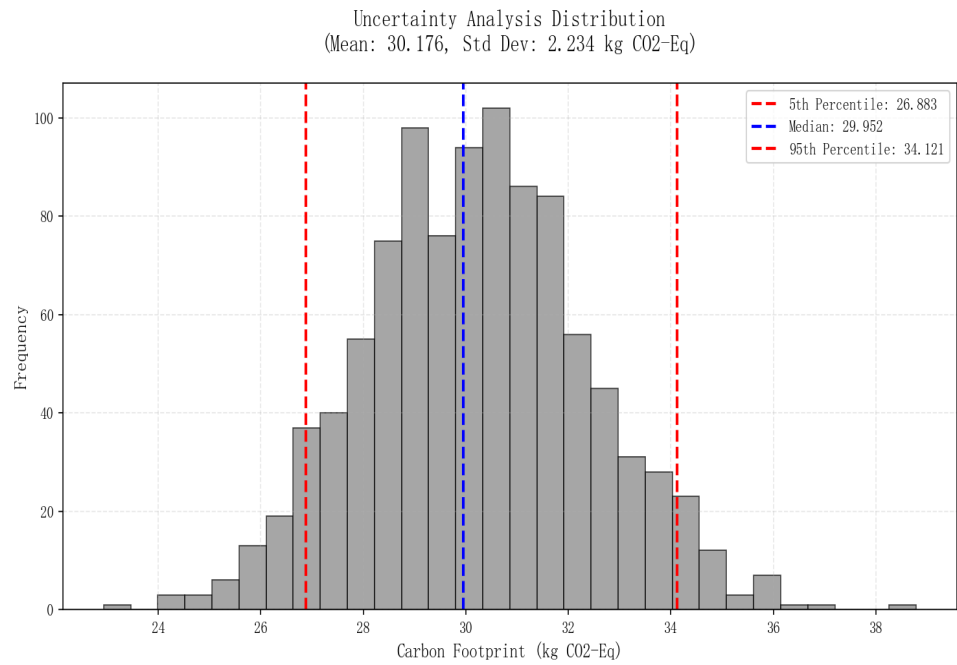
Life Cycle Stage	Carbon Footprint (kg CO ₂ -Eq)	Percentage (%)
Raw Material Acquisition	1.202	4.00
Production	5.380	17.89
Distribution	0.588	1.95

Use	20.264	67.40
End of Life	2.632	8.76
Total	30.066	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO2-Eq)
Mean	30.176
Standard Deviation	2.234
Coefficient of Variation (CV)	7.40%
Median	29.952
5th Percentile	26.883
95th Percentile	34.121

VI. Interpretation of Results

1. Description of Results

The total carbon footprint of the keyboard is 30.066 kg CO₂-Eq, a moderate value for an electronic input device. This result is dominated by the use phase, which accounts for 67.4% (20.264 kg CO₂-Eq) of the overall impact. Such a high contribution is typical for products with a prolonged operational lifetime and significant electricity consumption, as the keyboard's energy use during typing and backlighting (if applicable) accumulates over years of service. Production contributes 17.9% (5.380 kg CO₂-Eq), indicating that manufacturing processes and component assembly—such as plastic molding, circuit board fabrication, and electronics—are the second major hotspot. Raw material acquisition (4.0%, 1.202 kg CO₂-Eq) and distribution (2.0%, 0.588 kg CO₂-Eq) are relatively minor, while end-of-life treatment (8.8%, 2.632 kg CO₂-Eq) shows a significant footprint from disposal, likely due to energy-intensive recycling or incineration of mixed materials. Overall, the use stage is the primary emission driver, with production also warranting attention for reduction strategies.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a typical user scenario with average daily usage hours and a standard electricity grid mix, potentially representing a country average. The use-phase emissions are calculated based on a power consumption estimate of the keyboard (wired or wireless, with a battery) and an assumed lifespan of 3–5 years. Distribution emissions assume a regional transport distance and standard logistics. Data limitations are evident from the “Unknown” entries in top process contributions, indicating that process-level emission factors for specific components (e.g., keycaps, switches, cables) are aggregated or unavailable. The lack of detailed material and energy flows for sub-assemblies reduces granularity and may obscure minor but impactful processes. System boundaries follow a cradle-to-grave approach, yet they exclude capital goods, maintenance, and software-related impacts. The 90% confidence interval (26.883–34.121 kg CO₂-Eq) from the Monte Carlo simulation reflects moderate uncertainty (CV 7.4%), driven by variability in user behavior and end-of-life scenarios.

3. Suggestions for Improvement

- To reduce the carbon footprint of the keyboard, the following specific improvement measures are recommended:
- ****Reduce use-phase energy consumption****: Implement low-power components (e.g., energy-efficient microcontrollers and backlight LEDs) and enable automatic sleep modes after short idle periods. Encourage users to pair the keyboard with renewable energy sources (e.g., solar chargers for wireless models) or carbon-aware usage patterns.
- ****Optimize production processes****: Substitute virgin plastics with recycled or bio-based polymers in keycaps and housing, and adopt cleaner manufacturing technologies (e.g., injection molding with lower energy intensity). Prioritize suppliers with certified low-carbon production and renewable energy usage.
- ****Improve end-of-life management****: Design for disassembly to facilitate separation of materials (e.g., metal contacts, plastic parts, and electronics), and establish take-back programs to ensure high recycling rates rather than landfill or incineration. Use modular designs to extend product lifespan.

- ****Enhance data quality****: Replace aggregated “Unknown” process contributions with primary data from suppliers on specific components (e.g., integrated circuits, cables). Conduct detailed process-level LCAs for raw material extraction (e.g., plastics, metals) to refine uncertainty bounds and identify hidden hotspots.